

Resource Riches, Rebel Rulers: The Impact of Resource Wealth on Rebel Governance

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Abstract: Does natural resource lootability influence the development and maintenance of rebel governance? Are some kinds of governance more likely than others to develop when groups are financed by natural resources? A growing literature examines how natural resources influence non-violent tactics during civil conflict, such as rebel governance. This study contributes to this literature by investigating how the lootability of natural resources shapes incentives surrounding the development of security institutions and social services. I argue lootable resources create incentives to develop social services, while non-lootable resources promote the development of security institutions. I find evidence that the type of natural resource matters: groups with only non-lootable resources are more likely to develop security institutions than those with lootable resources or those with no resources. Further, groups with lootable resources are more likely to develop social services than those with no resources at all. This study highlights that investigating the incentives surrounding natural resources as a source of capacity can prove insightful for understanding variation in group behavior.

Introduction

The Islamic State – West Africa Province (ISWAP) splintered from Boko Haram in 2016; quickly thereafter, it subsumed much of Boko Haram's membership, growing in military power through a series of victories (Stoddard, 2023). The major difference that promoted ISWAP's growth and Boko Haram's corresponding decline was ISWAP's development as an alternative to the state and greater focus on attacks towards military and police targets rather than violence against civilians (Sempijja et al., 2023). ISWAP's expansion and entrenchment in the Lake Chad region, ideal land for agriculture and fishing, granted it access to consistent resource wealth from taxation from fisherman, livestock owners, and farmers (Samuel, 2025; Foucher, 2020), something Boko Haram lacked (van der Merwe, 2017). This wealth created a number of opportunities for the organization to expand its footprint.

ISWAP's wealth enabled it to develop several free services and institutions such as healthcare, potable water, and latrine development. Additionally, it supported institutions that furthered profits and thus maintained its larger capacity for war; providing free seeds and pesticides (as long as resulting produce was sold only inside ISWAP territory), boats and canoes for transportation, grain-grinding machines, and interest-free loans for traders (Samuel, 2025). ISWAP's ability to meaningfully control trade, taxation, and agriculture in resource-rich territories helped sustain the organization's ability to engage in governance alongside its broader conflict efforts (Nyelade, 2025). ISWAP's growth has given it influence over the populace and economy in various localities in northern Cameroon, eastern Niger, and northeastern Nigeria (Barkindo, 2023), with a heightened emphasis on control over people rather than territory alone to bolster its resilience to counterinsurgency efforts (Stoddard, 2023). ISWAP's use of

governance alongside its resource wealth raises two related questions: do qualities of rebels' access to natural resources matter for institutional development? Are certain kinds of governance more likely than others to develop when groups are financed by natural resources?

Extensive research has highlighted a variety of circumstances in which natural resources can influence civil conflicts and rebel group behaviors, oftentimes at the detriment of local populations (Whitaker et al., 2019; Laurent-Lucchetti et al., 2024; Haer et al., 2020; San-Akca et al., 2020; Lujala, 2009; Lujala, 2010). At the same time, a wealth of literature showcases how rebel governance is influenced by, and in turn influences, a variety of dynamics in civil conflicts (see, for instance, Arjona, 2016; Arjona et al., 2015; Mampilly, 2011; Cunningham & Loyle, 2021; Loyle et al., 2023). Thus, while work on natural resources has extensively covered how resource presence, access, type, and profit method influences decision-making regarding violence, a similar look at how natural resources impact rebel governance is needed to better understand how natural resources influence non-violent rebel behavior. I build on existing work that has begun such investigation (e.g., Mampilly & Gutierrez, 2025; Conrad et al., 2022; Yalaz et al., 2026; Soules, 2025a) by providing evidence that the *lootability* of natural resources creates direct incentives for groups to provide social services, highlighting how the qualities of natural resources are important to explain variation in non-violent behaviors as well as violent ones.

By addressing this gap, this paper contributes to the expansive body of work that has examined the influence of natural resources on state governance practices (most notably through an examination of the “resource curse” (Badeeb et al., 2017; Brunnschweiler & Bulte, 2009; van der Ploeg, 2011; Ross, 1999; Ross, 2015; Rosser 2006; Sala-i-Martin and Subramanian 2013)). Importantly, the effect of the resource curse on state governance varies depending on the nature of the resource and incentives of the state (Ross, 2015). I argue that the nature of the resource

(e.g., whether it is lootable) also shapes rebel governance, influencing decision-making over what governance institutions to prioritize.

Establishing and maintaining governance produces clear benefits, such as information, legitimacy, and public support (Florea and Malejacq, 2023). Engaging in governance is costly; natural resources can help offset the cost of governance, however. Security-related institutions support existing economic capacity. Social services build more traditional implements of governance that can, for example, help secure the support of locals, even if such institutions are more costly than other forms of governance. Having lootable resources makes the development of social services more desirable. Groups can leverage the resources' relative ease of access to reduce the costs of social services, reducing the groups' barriers for governance provision. Additionally, groups with lootable resources may see unique value from social services such as education that improve intragroup cohesion due to a greater influx of lower quality recruits that are drawn to easier resource profits (Weinstein, 2005). Having non-lootable resources incentivizes the development of security institutions. Because investment to develop the resource is more costly, institutions that uphold and support a consistent presence in the locality are more valuable.

I find empirical evidence that natural resources shape the types of governance rebel groups provide. This advances the literature by providing further evidence that natural resources change group incentives and encourage the development of non-violent strategies. Groups financed by natural resources are more likely to develop governance institutions. Lootable resources particularly incentivize the development of social services compared to groups with no resources, with some evidence that even among groups with resources, those with lootable resources are particularly likely to develop social service institutions. Additionally, groups with

non-lootable resources are more likely to develop security institutions than those with lootable resources. By better understanding the incentives surrounding natural resources' value towards governance, scholars and policymakers can better anticipate how changes in resource access and value will affect group behavior as an alternative to the state.

Literature Review

Early research on the effect of natural resources showcased how the mere presence of resources in a conflict can lead to more severe conflicts, whether it be from gems (Lujala et al., 2005; Lujala, 2009), oil (Lujala et al., 2007; Lujala, 2009), or gas (Lujala, 2010).¹ As the field has grown, greater attention has been paid to the specific incentives that natural resources create to influence conflict processes. For instance, access to natural resources creates conditions where violence against civilians is less restrained and conflict is worsened (Whitaker et al., 2019; Laurent-Lucchetti et al., 2024; Haer et al., 2020; San-Akca et al., 2020; Lujala, 2009; Lujala, 2010; Fortna et al., 2018). Rebel groups with natural resources often become less reliant on the local population than rebel groups without. Rather than rely upon locals for their needs, easy access to resources can incentivize rebels to use greater levels of violence against civilians and create conditions where alienating civilians is more permissible (Whitaker et al., 2019). However, other scholarship finds that this relationship is dependent upon the strategies taken by the group and the nature of the resource itself.

Rebels are strategic in how natural resources influence their tactics. For example, international efforts to regulate the illicit diamond trade shifted violent conflict towards substitute resources like tin, tungsten, gold, and more (Saab, 2025). Further, Whitaker et al.

¹ Drug cultivation, conversely, can result in lower conflict intensity (Lujala, 2009), despite its ability to strengthen relative rebel capacity (Cornell, 2007; Cornell, 2005).

(2019) find that rebels that *extort* natural resources engage in more sexual violence than groups that *smuggle* them because each strategy creates different constituencies that groups rely on. Likewise, groups are more likely to engage in violence when resource prices fluctuate, depending on whether extraction is labor-intensive or capital-intensive (Laurent-Lucchetti et al., 2024). Rebels are shown to be strategic actors, and that qualities of natural resources subsequently shape their strategies towards violence. This paper furthers findings such as these by showcasing how natural resource lootability incentivizes changes to rebels' non-violent strategies, particularly surrounding the development of their own institutions.

A growing literature has examined the role of rebel governance and its impact on civil conflict (Albert, 2022; Florea & Malejacq, 2023; Hassan, 2022; Park, 2024; Teiner, 2022). Rebel governance is conceptualized here as "A set of actions insurgents engage in to regulate the social, political, and economic life of non-combatants" (Albert, 2022, p. 623). Recent work has also begun to examine the effect of natural resources on governance provision in cross-national contexts (Conrad et al., 2022; Yalaz et al., 2026). This paper expands on these projects in two primary ways.

First, this paper highlights motivations behind the use of natural resources to finance governance beyond labor; while this is a key aspect, fungible capital like resource wealth can be used for several aims such as improving internal cohesion or developing legitimacy to support groups' larger objectives. Second, rebel groups vary extensively in their levels of organization and governance provision (Huang, 2016). For instance, in Angola, the National Union for the Total Independence of Angola (UNITA) featured a president, congress, committees, a constitution, and diplomatic efforts (Huang, 2016). In contrast, ISWAP has engaged in taxation, Sharia-based justice (with brutal punishment), security, and dispute resolution mechanisms

(Hassan, 2022). Across groups that provide governance, the nature of that governance varies widely.

This paper provides evidence that differences in governance provisions depend on the needs of the organization. Further, this paper reinforces that sources of economic capacity affect groups' behavior. In this case, the (non)lootability of natural resources creates incentives for groups to develop specific types of governance: security institutions and social services.

Natural Resources, Lootability, and Rebel Institutions

I argue that natural resources aid in the development and maintenance of rebel governance. While rebels can acquire a variety of benefits from rebel governance, governance is a costly endeavor, with groups facing financial burden from monitoring/enforcement and goods-provision (Florea and Malejacq, 2023). Engaging in services like education, the development of healthcare, or establishing a police force can require significant investment, especially if governance is new for the organization. Natural resources provide rebel groups with the financial capacity to engage in governance via fungible profits, akin to the ways in which natural resources offer a means to meet the logistical requirements for insurgency (Weinstein, 2005).² With this enhanced capacity, groups are capable of expanding their operations to adopt new tactics or otherwise innovate on current strategy.

² A similar parallel has been drawn between external support and rebel governance. Revenue sources such as external support can help supplement the costs of insurgency, allowing rebels to divert resources to institutions like education and healthcare (Huang & Sullivan, 2021). Natural resources and external support share some parallels that help support the plausibility of natural resources as a finance tool for governance. Like natural resources, external support can reduce dependence on a local population (Salehyan et al., 2014). Further, external support can be fungible like natural resource profits, giving rebels flexibility in its usage (Meier et al., 2023; Huang & Sullivan, 2021).

Developing governance institutions provides several benefits which enhance the payoff from natural resource wealth. When groups are profiting from natural resources, engaging in governance maintains a consistent presence where their resource wealth is located, enabling greater extraction. Additionally, groups can develop institutions to invest in human capital, improving the quality and condition of workers and soldiers. Even outside of extracting a greater return on investment, groups can use natural resource wealth to directly derive other benefits of governance, such as legitimacy; by doing so, groups can transform some of their economic capacity into greater political capacity. These mechanisms are reflected empirically in both the cases of the Kachin Independence Organization (KIO) in Myanmar and the Allied Democratic Forces (ADF) in Uganda. I briefly describe the background for each case below before tying them to each mechanism.

During the 1990s, the KIO was an active faction in northern Myanmar (Brenner, 2017). The KIO agreed to a ceasefire in 1994, after pressure by a government offensive left the KIO severely weakened. In Myanmar, ceasefire agreements with groups like the KIO included giving the group better access to natural resource production and the resulting profits as a means to co-opt leadership via economic incentives (Brenner, 2015; Venning, 2019). For the KIO, a ceasefire thus provided an opportunity to recuperate and govern territory within Kachin state, while also providing a powerful opportunity to expand the organization through resource wealth (Brenner, 2015).

The ADF originally formed in eastern Democratic Republic of the Congo (DRC) in 1995 to wage war against the Ugandan government (Thompson, 2021). The Sudanese government played a direct role in their formation (Day, 2011), and the ADF had initially received direct support in terms of training and logistics from the Sudanese government (Thompson, 2021).

However, by 2001, the ADF had degraded in strength to the point where they were declared defeated in 2007 (Day, 2011; Thompson, 2021). During this time, the ADF chose to remain dormant in the eastern DRC, reducing its attacks for nearly a decade from 2003 to 2012 (Thompson, 2021). Further, the ADF avoided public proclamations, social media, and punished people trying to leave, tightly controlling movement surrounding their camps, and minimizing activity that could draw attention to their activities and objectives (Titeca & Fahey, 2016). During this dormancy, the ADF shifted attention towards managing and extracting profits from the remote, rural territory in its possession (Cullen, 2017).

Natural Resources and Security

Among immediate priorities for new groups, survival is paramount. Wealth from natural resources can provide a valuable gateway for rebel groups to maintain relevancy and develop their capacity for war. For the ADF, a shift towards commercial interests began as soon as they retreated from Uganda into the eastern DRC in 1995. Rather than rebuilding and seeking a renewed war immediately, the ADF entrenched themselves in local black markets, cross-border smuggling, and patronage systems, venturing into timber, gold mining, coffee, and marijuana markets (Thompson, 2021; Cullen, 2017; Titeca & Vlassenroot, 2012). Where it was once a means to finance their objectives, profits later became their primary goal (Cullen, 2017). During dormancy, the ADF sought to develop commercial activities and operate legitimate businesses (Cullen, 2017). Once the ADF had established itself as a firm alternative to the state, it could leverage its control and enhance their conflict presence.

As wealth accumulated, the ADF's base of resources became more stable, increasing their capacity to fight the DRC; by using their extensive commercial network which featured a diversified set of resources, the ADF improved their resiliency to absorb losses and market

downturns (Reeder et al., 2023). Additionally, the ADF maintained regular relations with local businesses and political leaders, and contacts with both national and international financiers with the nominal goal of overthrowing the Ugandan government (Nsobya, 2018). As rebellion enabled the ADF to profit, it was then able to reinvest its resource wealth into stability through the development of security institutions, consequently creating the means to grow more relevant both economically and politically. At the same time, organizations can leverage security institutions to defend and further scale their natural resource operations.

Security institutions, I argue, create the means to more effectively and efficiently extract profits from extortion, taxation, and smuggling. For instance, the ADF operate mines, sell licenses for tree felling operations, smuggle consumer goods from Uganda into the DRC to fill a supply gap for products like coffee and pharmaceuticals or fuel. Members of local authorities are complicit in their activities, which allows them to operate these sites as well as use transportation routes (Cullen, 2017). For the KIO, security institutions could be leveraged to further secure and garner resource wealth (Gluckman, 2015; Combs, 2021). Developing such institutions helps delineate boundaries of influence, particularly relevant for groups seeking autonomy or independence, while simultaneously creating a means to finance its existence through greater involvement in managing natural resource extraction and smuggling. Otherwise, groups may seek to establish a border patrol as a way of ensuring that resource wealth that is crossing their claimed region is providing economic value to the organization.

Further, security-related institutions serve to establish coercive influence over the local populace. Developing security-related institutions such as immigration and border control (as in the case of the KIO) or prisons, internal security, and police (as in the case of the ADF) helps defend existing profits through more effective enforcement and coercion of the local populace.

For an organization to manage and establish bureaucracy over resource extraction, the organization needs reliable and credible means of punishing defectors. While regular soldiers are more than capable of doing so, groups in the long term may seek to develop internal security forces near such localities or along smuggling routes to regulate the resource economy more effectively.

Natural Resources and Social Services

An important deterrent of the development of institutions is the financial cost (Florea and Malejacq, 2023). Security institutions such as a border patrol have financial costs but are not far removed from costs a rebel organization typically faces (e.g., personnel, weapons, outposts, etc.). Social services, however, can come with new costs (e.g., curriculum, administrative staff, founding schools, libraries, and clinics, establishing programs related to redistribution, licensure, or services such as vaccinations, etc.). Roads, managing schools, providing medical care, housing and transportation, therefore, require heavy financial investment to initiate programs. Building necessary infrastructure, providing wages for workers, and creating incentives for participation all take capital. Natural resources can provide immense wealth to finance such efforts. From 1988 to 2012, foreign direct investment (FDI) surged into the resource-rich border provinces where the KIO resided, with jade exports (primarily from Kachin state) worth anywhere from \$6 to \$9 billion USD in 2011 alone (Brenner, 2015). As a result, the development of less essential services (i.e., services that do not directly uphold profit) will become more feasible even if they are further outside a group's initial expertise or have higher costs than other forms of governance.

Furthermore, organizations that want to maintain continuous profit from natural resources must reinvest in the community to do so. Mining entails a more intensive reliance on local

populations; those local populations are also bearing more risk to extract the resource. Thus, the rebels need to provide healthcare to incentivize locals to collaborate with the rebels and minimize labor problems in the form of sick or injured workers, harming profits (Conrad et al., 2022). In the case of the KIO, working in the jade mines is especially dangerous for those extracting and creates considerable pollution and environmental damage at the same time (Roi, 2022). Similarly, local government officials in the region directly supported the development of infrastructure to facilitate resource extraction and border trade, enabling easier timber extraction, drug trafficking, and jade mining (Woods, 2011; Brenner, 2015). Developing social services such as healthcare or infrastructure thus provides a way of maintaining or growing profits from natural resources, despite higher upfront costs.

While natural resources create the financial capital to invest in social services, choosing to develop social services helps diversify the return on their investment into other forms of capacity. Even if further profits are not the main goal behind developing governance, the wealth that resources provide helps groups provide public goods, developing domestic legitimacy for the organization. As part of its dormancy, the ADF leveraged social services to create an independent community and social code among members (Cullen, 2017). Enhancing domestic legitimacy from public goods such as education can be intrinsically valuable, as in the case of the KIO.

Separatist groups or *de facto* states, in particular, need domestic and international legitimacy; thus, public goods and governance is an important priority to allocate resources towards (Yun & Ishiyama, 2025). The KIO has a number of institutions, including schools, immigration and border control, diplomacy, and a judicial system. These institutions were supported by the extraction and export of jade and timber to China (Venning, 2019; Brenner,

2015). Natural resources provided the means to engage in governance practices that an autonomy-seeking group like the KIO is directly incentivized to pursue as a prerequisite to achieve their objectives: social services and border delineation through security (Yun & Ishiyama, 2025; Stallman & Griffiths, 2024). At the same time, natural resource profits that improved the group's economic capacity created legitimacy issues for the KIO surrounding concerns of corruption and wealth inequality (Aung, 2025; Libre, 2009); providing governance can also serve as a means to maintain the larger benefits of resource wealth while minimizing the costs to fundamental resources like legitimacy. For groups to be able to establish themselves as a meaningful alternative to the state, financial resources to provide a diverse set of quality social services are important; natural resources serve as one such avenue to do so.

How Lootability Shapes Incentives

While it is clear natural resources provide greater means to develop different institutions, I argue the *nature* of the resource in question ought to be considered as well. While natural resources increase rebel groups' economic capacity, the resource itself creates distinct incentives and priorities for how the group spends that income. Specifically, the lootability of the resource shapes whether the group will prioritize security or social governance.

Lootable resources consist of particularly valuable resources in a market with low barriers to entry (Findley & Marineau, 2015); loutable resources are often simple to harvest with a low burden for labor and greater ease of smuggling (Lujala et al., 2005). By harvesting loutable resources, groups can profit more quickly compared to non-loutable resources (Ross, 2004; Snyder, 2006). Further, because of this relative ease of access, groups face unique pressures that incentivize them to invest in social services.

I argue that lootable resources will encourage the development of social services relative to non-lootable resources in two ways. First, lootable resources' ease of access enables groups to leverage that quality into direct service provision. For instance, the FARC-EP put 50% of its revenue from coca back into local communities (Mampilly & Gutierrez, 2025). The KIO was able to leverage taxes on new mining companies, and often developed infrastructure in coordination with mining companies, as they could now exploit natural resources in their territory at unprecedented levels during the ceasefire (Brenner, 2015). Similarly, the KIO sold logging rights in exchange for the development of two hydropower plants to improve electricity access (Brenner, 2015). The ease of profit from lootable resources enabled the KIO to leverage this quality to share the costs of service development with external corporations. These quick profits and direct exchanges help groups resolve the challenges of high capital need for service provision.

Second, this enhanced accessibility for profit may exacerbate the dilemma of natural resources attracting less disciplined/more opportunistic recruits, reducing the quality of recruits (Weinstein, 2005; Haer et al., 2020; Aronson et al., 2024; Sobek & Thies, 2015; Lujala et al., 2005). However, a plethora of recruits from natural resources may make providing services that are intensive for human capital, such as a police force, infrastructure development, or managing a justice system, more viable given the expanded membership within the organization. Further, if lootable resources attract lower quality recruits, these resources may also spur the development of institutions that help *manage* these recruits. If profits are slower to obtain or face more barriers such as with non-lootable resources, the development of institutions as management is more difficult.

Social services can also serve to improve group cohesion. Education can be used as a tool to socialize civilians and develop ideological appeals that promote recruitment and survival (Palik et al., 2025). Intragroup ideological diversity spurs internal fractionalization and commitment problems which further lengthens the duration of the conflict (Soules, 2025b), while better socialization helps ensure internal cohesion and discipline (Oppenheim & Weintraub, 2017; Hoover Green, 2018). Developing institutions that encourage cohesion will be more valuable for organizations with lower quality recruits.

For instance, the FARC's education services included specialized teachings related to tactics and political philosophy as well as military training (Placidet, 2025; Lizarazo-Vega, 2021), while instilling a collective identity in line with FARC's ideology (Placidet, 2025). FARC's educational services were directly tied to their larger objectives, while still providing a public good in response to local needs (Lizarazo-Vega, 2021). Education thus helps develop norms and rules both for civilians and materially driven recruits to promote long-term group survival. Educational services can therefore serve as a means of non-violent intragroup socialization to help address the issues that natural resources, and especially lootable resources, create for recruitment efforts. I thus derive the following hypothesis:

Hypothesis 1: Rebel groups that profit from lootable natural resources will be more likely to develop social services than groups profit from non-lootable natural resources and groups that have no resources.

Non-lootable resources, such as oil and ores, do not have the same degree of accessibility as lootable resources. This incentivizes the development of security institutions to maintain consistent access to the locality. Because non-lootable resources are characterized by being more difficult for smuggling and having higher labor costs, security becomes a more immediate

priority. Investment into formalized institutions dedicated to security serves as a useful means to legitimize practices such as protecting mining sites in exchange for taxation (Krauser, 2020), for instance. Further, resource extraction for non-lootable sites requires a greater investment of resources, such as labor and the necessary infrastructure, meaning that the costs of losing access to the locality are more stark.

Security institutions may help with maximizing wealth gains from natural resources as well. To more effectively extort or smuggle resources that require a greater degree of investment to reap profits, developing institutions that help delineate presence and create natural checkpoints for collecting fees or monitoring access to the region is ideal. For instance, the KIO's efforts to engage in rare earth mining result in companies engaging in payments for permissions and security guarantees for mining activities, including safe passage along roads or checkpoints to the border, while workers from China register with an immigration office (Meehan & Lawn, 2024). As a result of more intensive resources that are often needed to be dedicated to resource extraction, developing institutions that make collaboration with external parties while maintaining a sufficient armed presence more feasible is key. This leads me to my second hypothesis:

Hypothesis 2: Rebel groups that profit from non-lootable natural resources will be more likely to develop security institutions than groups profit from lootable natural resources and groups that have no resources.

While this theory neatly separates groups into those with lootable and non-lootable resources, groups oftentimes have access to both. In the sample analyzed below, for instance, there are no groups with only lootable resources; groups with access to lootable resources also have access to non-lootable resources. In the event groups have access to both forms of natural

resources, I expect that their propensity for security institutions to be lower than groups with non-lootable resources, while the propensity for social services to be higher than groups with non-lootable resources. While the flexibility of lootable resources and the diversification from multiple resource types can grant greater financial resources, having access to both creates dual pressures to develop social services and security institutions. I anticipate groups that profit from both forms of resources will prioritize social services, given that institutions such as healthcare and infrastructure benefit the development of profit for both forms of resources.

Research Design

To assess my hypotheses, I construct a dyad-year dataset for all conflict and inter-conflict-years, based on the UCDP Dyadic Dataset version 24.1 (Davies et al., 2024; Harbom et al., 2008). The first year a dyad is included is the first year of active conflict or 1990, whichever is later. The last year a dyad is included is the final year of active conflict or 2012, whichever is earlier. Because groups can profit from natural resource wealth and develop institutions even without actively fighting the state, if a dyad has a gap between two active conflict periods, those interim years are included in the sample. This produces a sample of 783 dyad-years. Descriptive statistics for all included variables can be found in the Appendix.

For my dependent variables, I rely upon the data from the Rebel Quasi-State Institutions dataset (Albert, 2022). This dataset contains twenty-five different indicators for whether a rebel group provided a certain type of rebel governance, ranging from political parties to media outlets to border patrols. I rely on Albert's (2022) conceptualization of security and social service-related institutions and create two measures for each (see Table 1). First, a binary measure for when rebel groups have at least one institution of that respective type (1) or not (0). Second, a count measure for the number of institutions that a rebel group has within each respective type.

Table 1: Categorizations of Rebel Quasi-State Institutions

DV Classification	Rebel Quasi-State Institutions Indicators (Albert, 2022)
Security-related Institutions	Border patrol, IDs, armed forces, “other military”
Social Services	education, healthcare, infrastructure, transportation, policing, law, justice, housing, constituency politics, "other social"

For my key independent variable, I rely upon the Rebel Contraband Dataset (Walsh et al., 2018), which provides information on the resources and methods for profit from said resources that rebel groups used over time. I code rebel groups as a 0 if they do not profit from any *natural resource* in a given year (used as the baseline category), a 1 if they only profit from non-lootable resources, and a 2 if they profit from at least one lootable resource. For my list of lootable resources, I look to Findley and Marineau (2015) and include the following resources: alluvial diamonds, drugs, timber, and gems. While theoretically rebel groups could have exclusively lootable resources in a given year, no such cases are present in my sample.

To help account for potential endogeneity and because natural resource access does not immediately translate into wealth and the potential for institutional development, I lag the natural resource variable by one year. For robustness, I examine a non-lagged alternative, where groups in the sample have no resources, non-lootable resources, only lootable resources, or both, and use two-way fixed effects (TWFE) to help address potential endogeneity from the non-lagged alternative measure and find evidence that social service provision is particularly impacted by resource type.

To address potential confounders, I include a series of controls. At the group-level, I include whether groups control any territory, an ordinal scale of the relative strength of the rebel

group compared to the government, and an ordinal scale for how centralized the rebel group is, all sourced from the Non-State Actor (NSA) Dataset (Cunningham et al., 2009). I anticipate that each of these variables should increase the likelihood of rebel governance, as groups that are stronger, more centralized, and control territory are in a better position to invest in governance (Conrad et al., 2022; Florea and Malejacq, 2023). I include binary variables for whether rebel groups are founded with an objective of independence or autonomy and if they are founded with an ethnic identity, sourced from the Foundations of Rebel Group Emergence (FORGE) Dataset (Braithwaite & Cunningham, 2020). I also include whether rebel groups are founded with any ideology as groups from diverse ideologies share similar incentives to develop governance (Mampilly & Stewart, 2021).

At a country-level, I include variables to capture a country's V-DEM polyarchy score (Coppedge et al., 2011, Coppedge 2021), GDP per capita (World Bank, 2024 -a), and population (World Bank, 2024 -b). At the conflict-level, I include a count of the number of external supporters involved in the conflict in a given year using the UCDP External Supporters Dataset (Meier et al., 2023) given that external support could influence rebel governance and groups that use natural resources may be less likely to receive external support (Huang & Sullivan, 2021; Salehyan et al., 2011). Further, I include whether the conflict is "inactive" (between 0 and 25 battle-related deaths in a given year), "minor" (25-999 battle-related deaths) or "major" (1,000 or more battle-related deaths). Lastly, I include region-level fixed effects to account for any other potential confounders that may exist at a larger level.

Analysis

I estimate logistic (see Table 2) and negative binomial (see Table 3) regressions with robust standard errors clustered by country.

In each table, I assess Hypothesis 1 in M1 and M2; M1 sets the baseline set to those with no resources, while M2 sets the baseline to those with only non-lootable resources. I assess Hypothesis 2 in M3 and M4, with the same respective baselines. The results are discussed below.

Table 2 - Rebel Governance Provision

	M1: Security Institutions (H1)	M2: Security Institutions (H1)	M3: Social Services (H2)	M4: Social Services (H2)
No Natural Resources		-3.499* (0.816)		0.140 (1.567)
Only Non-lootable Resources	3.499* (0.816)		-0.140 (1.567)	
1 or More Lootable Resources	1.532* (0.462)	-1.967* (0.573)	1.923* (0.421)	2.064 (1.410)
Territorial Control	0.741* (0.251)	0.741* (0.251)	0.708* (0.209)	0.708* (0.209)
Strength of Centralization	1.416* (0.624)	1.416* (0.624)	1.708* (0.547)	1.708* (0.547)
Relative Rebel Strength	0.845 (0.489)	0.845 (0.489)	0.662 (0.446)	0.662 (0.446)
Independence/Autonomy Movement	3.970* (0.926)	3.970* (0.926)	2.134* (0.535)	2.134* (0.535)
Founded w/ Ideology	0.244 (1.549)	0.244 (1.549)	0.085 (1.029)	0.085 (1.029)
Founded w/ Ethnic Identity	-2.012* (0.998)	-2.012* (0.998)	-1.746* (0.740)	-1.746* (0.740)
External Supporters	-0.024 (0.098)	-0.024 (0.098)	0.318* (0.109)	0.318* (0.109)
Polyarchy	-0.976 (2.059)	-0.976 (2.059)	-1.176 (2.448)	-1.176 (2.448)
GDP per capita (thousands)	-0.224 (0.145)	-0.224 (0.145)	0.084 (0.075)	0.084 (0.075)
Population (millions)	0.001 (0.001)	0.001 (0.001)	0.003* (0.001)	0.003* (0.001)
Intensity Level	0.428 (0.287)	0.428 (0.287)	0.479 (0.306)	0.479 (0.306)
Constant	-8.311* (3.159)	-4.812 (2.768)	-7.163* (2.536)	-7.303* (2.672)
Region Fixed Effects	✓	✓	✓	✓
Observations	783	783	783	783
Pseudo R^2	0.410	0.410	0.389	0.389

Standard errors in parentheses and clustered by country

No resources is the baseline in Models 1 and 3. Only having non-lootable resources is the baseline in Models 2 and

4.
* $p < .05$

The results indicate support for Hypothesis 1. I find that groups with non-lootable resources are more likely to develop security institutions compared to groups that do not have access to some form of natural resource profit. When calculating the predicted probabilities, I find that groups with only non-lootable resources are nearly ten times more likely to have a security institution than those with no resources at all.³ In addition, I find that groups with both resource types are less likely to have security institutions than those with only non-lootable resources. Groups with no resources are less than half as likely (with a predicted probability of 5.3%) as those with lootable resources (16.1%) to develop security institutions. However, groups with only non-lootable resources are three times as likely to have security institutions than those with both lootable and non-lootable resources.

When investigating Hypothesis 2, I find conditional support for Hypothesis 2 for groups with both lootable and non-lootable resources. Groups with only non-lootable resource profits are not more likely to provide social services than groups with no resource profits at all. Calculating the predicted probabilities shows almost no difference in the probability of having at least one form of social services between those with no resources (9.7%) and only non-lootable resources (8.7%). Groups with both forms of resources are over three times as likely to have at least one form of social services, with a predicted probability of 35.6%. However, I also find no significant difference between groups with non-lootable and both types of resources when it comes to social service provision. This may be because what constitutes social services is a

³ Predicted probabilities are calculated with ordinal variables held at their median values, binary and nominal variables at their mode, and continuous variables at their mean.

broader categorization than security institutions. For instance, when assessing individual social services, I find groups with both types of resources are significantly more likely to provide education and healthcare, but not services like policing and justice which also serve security-related functions.

Table 3 investigates whether resource access influences the *number* of different institutions of each type that groups provide. I again find that both resource types positively influence the development of security institutions, while only groups with lootable resources will have a higher count of social services than groups with no resources at all, supporting the conditional results of Hypothesis 2 found earlier. When calculating the predicted counts, groups with non-lootable resources see an over 300% increase in their predicted number of security institutions compared to those with no natural resources. Those with lootable resources see a 79.6% increase in their predicted number of security institutions. When examining differences between groups with resources by type, I again find support for Hypothesis 1. Calculating the predicted counts shows groups with only non-lootable resources have double the predicted number of security institutions than those with both resource types. However, while in the expected direction, I find no significant difference between groups with non-lootable and lootable resources when it comes to social service provisions.

Table 3 - Count of Rebel Institutions

	M1: Security Institutions (H1) (Count)	M2: Security Institutions (H1) (Count)	M3: Social Services (H2) (Count)	M4: Social Services (H2) (Count)
No Natural Resources		-1.452* (0.352)		0.371 (0.837)
Only Non-lootable Resources	1.452* (0.352)		-0.371 (0.837)	
1 or More Lootable Resources	0.584* (0.271)	-0.867* (0.268)	0.840* (0.213)	1.211 (0.722)
Territorial Control	0.375* (0.098)	0.375* (0.098)	0.533* (0.097)	0.533* (0.097)
Strength of Centralization	0.553* (0.216)	0.553* (0.216)	1.054* (0.226)	1.054* (0.226)
Relative Rebel Strength	0.541* (0.222)	0.541* (0.222)	0.319 (0.279)	0.319 (0.279)
Independence/Autonomy Movement	2.281* (0.455)	2.281* (0.455)	1.558* (0.425)	1.558* (0.425)
Founded w/ Ideology	0.039 (1.033)	0.039 (1.033)	0.538 (0.733)	0.538 (0.733)
Founded w/ Ethnic Identity	-0.771* (0.361)	-0.771* (0.361)	-1.579* (0.352)	-1.579* (0.352)
External Supporters	-0.034 (0.054)	-0.034 (0.054)	0.158* (0.055)	0.158* (0.055)
Polyarchy	0.270 (0.724)	0.270 (0.724)	-0.608 (1.003)	-0.608 (1.003)
GDP per capita (thousands)	-0.116 (0.062)	-0.116 (0.062)	0.045 (0.032)	0.045 (0.032)
Population (millions)	0.000 (0.001)	0.000 (0.001)	0.002* (0.001)	0.002* (0.001)
Intensity Level	0.122 (0.145)	0.122 (0.145)	0.073 (0.155)	0.073 (0.155)
Constant	-5.511* (1.242)	-4.059* (1.188)	-4.687* (1.498)	-5.058* (1.343)
Region Fixed Effects	✓	✓	✓	✓
Log(alpha)	-17.300* (0.476)	-16.617* (0.098)	-0.409 (0.604)	-0.409 (0.604)
Observations	783	783	783	783
Pseudo R^2	0.291	0.291	0.168	0.168

Standard errors in parentheses and clustered by country

No resources is the baseline in Models 1 and 3. Only having non-lootable resources is the baseline in Models 2 and 4.

* $p < .05$

When looking across controls, I find that more centralized groups and those with territorial control are more likely to have security institutions and social services across all models. Similarly, I find that groups seeking independence or autonomy are more likely to develop security institutions or social services. External support increases the likelihood of developing social services, but has no impact on group development of security-related institutions. Lastly, groups founded with an ethnic identity are significantly less likely to develop security or social service institutions.

To assess the robustness of these findings, I estimate additional models included in the Appendix. First, as mentioned before, I assess Hypothesis 2 with a set of models examining the provision of individual social services, examining the provision of education, healthcare, policing, and a justice system. I find that groups with lootable resources are significantly more likely than those with only non-lootable resources to have education or healthcare, but not policing or a justice system, indicating that within the broad category of social services, natural resource type incentivizes the development of different services. Second, I examine the effects of natural resources of any type compared to having no natural resources. I again find that groups with natural resources are more likely to develop security or social service-related institutions. Third, I include year-level fixed effects. I find the results do not meaningfully change with this decision. Fourth, I include country-level fixed effects. I find support for Hypothesis 2 between groups with non-lootable and both types of resources when limiting variation across countries; the results for Hypothesis 1 remains consistent.

Lastly, as an alternative model to address potential endogeneity between rebel resource access and rebel governance, I estimate TWFE models with the count of institutions of each type as the dependent variable. In the TWFE model, the effect of natural resources of either type on

security institutions is insignificant. The TWFE continues to find conditional support for Hypothesis 2. This may be due to the rarity of groups having multiple security institutions (less than 5% of dyad-years) compared to groups having multiple social services (approximately 25% of dyad-years), reducing the applicability of two-way fixed effects as a viable alternative design. When estimating a TWFE model with the binary dependent variables, effectively estimating Linear Probability Models (LPMs), I find the same pattern in regard to Hypothesis 1.

Discussion and Conclusion

Overall, I find support for my theoretical expectations. Security institutions are more likely to be present when groups profit from natural resources, especially from non-lootable resources. Further, I find evidence indicating that lootable resources incentivize the development of social services. These findings support a larger body of literature that showcases how the relative ease of lootable resources as a means to gain financial capacity can be especially useful for furthering the aims of the group or enabling tactics that do not rely on positive relationships with civilian populations (Haer et al., 2020; Aronson et al., 2024; Sobek & Thies, 2015; Lujala et al., 2005). Additionally, these findings align with suggestive evidence from the qualitative examination of the ADF and KIO.

This study demonstrates that natural resources do not just influence the provision of healthcare but rather amend incentives for the provision of multiple institutions. Further, these findings reify that better examining the nature of sources of rebel economic capacity is important to understanding their subsequent effect. Sources of capacity can incentivize different, seemingly opposite, forms of behavior; natural resources may spur violence, but they also spur the development of governance institutions, whether it be to enhance profit, further their larger objective, or improve their military capacity. Further insights into how natural resources

influence rebel governance create a more robust understanding of the role of natural resources and broader economic capacity in rebel group conduct. Similar to how the “resource curse” often creates problems for state governance, rebel organizations may suffer from corruption in leadership like states, as seen with the KIO. However, resource wealth can fuel the development of key institutions that can otherwise help groups achieve their aims while also providing benefits to civilians, making it less of a one-sided curse than one might believe.

While certain social services may become more likely when groups profit from lootable resources in particular, others are unaffected. As researchers continue to investigate the incentives to provide governance, evaluating a diversity of individual institutions within larger typologies may prove valuable to more robustly assess the potential simultaneous positive and negative effects on rebel tactics and decision-making within the broad category of governance. For instance, future research may seek to examine how the propensity to provide democratic institutions is influenced by natural resources, where groups may face a resource curse similar to states for these particular institutions (Sarkar & Sarkar, 2017; Marks, 2019; Ross, 2015). At the same time, further considering the role of group agency in providing governance through considerations of rebel objectives as a potential moderating factor on the influence of natural resources on governance is likely to provide further depth to this topic.

Future research should also continue to build upon these findings by examining how governments may engage in different forms of counterinsurgency as a result of rebel decision-making surrounding the use of natural resources as a funding source for governance. While natural resources may reduce rebel prohibitions surrounding violence against civilians, potentially triggering government violence of its own (Raleigh & Choi, 2017), the government may respond differently to organizations that use a mixed strategy of governance provision and

violence or, alternatively, prioritize governance over violence. Further qualitative analysis on a larger variety of cases than is presented here may also prove fruitful for more richly determining a causal pathway between governance and natural resources.

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Appendix

Table A1 - Descriptive Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Security Institutions	2224	.239	.427	0	1
Security Institutions (Count)	2224	.281	.541	0	3
Social Services	2224	.398	.49	0	1
Social Services (Count)	2224	1.28	1.94	0	8
Natural Resources (t-1)	1489	.433	.496	0	1
Natural Resource Usage (t-1)	1489	.79	.939	0	2
Natural Resource Usage	1474	1.029	1.13	0	3
Territorial Control	3465	1.365	1.494	0	3
Strength of Centralization	3155	2.075	.567	1	3
Relative Rebel Strength	3479	1.594	.666	1	5
Independence/Autonomy Movement	3250	.535	.499	0	1
Founded w/ Ideology	3343	.87	.337	0	1
Founded w/ Ethnic Identity	3343	.572	.495	0	1
External Supporters	2028	1.725	1.832	0	7
Polyarchy	3961	.324	.219	.007	.908
GDP per capita (thousands)	3306	2.677	7.128	.013	65.12
Population (millions)	3695	102.78	246.099	.397	1417.173
Intensity Level	4056	.912	.622	0	2
Region	4056	3.217	.969	1	5
Year	4056	1994.292	18.803	1946	2023
Country	4056	601.444	201.312	2	910

Table A2 - Individual Social Services

	M1: Education	M2: Healthcare	M3: Policing	M4: Justice System
No Natural Resources	1.281 (1.515)	0.525 (1.477)	-0.743 (1.606)	0.210 (1.272)
Both Lootable and Non-lootable	4.312* (1.252)	2.141 (1.240)	-0.684 (1.543)	0.928 (1.462)
Territorial Control	0.781* (0.225)	0.968* (0.287)	0.832* (0.295)	0.558 (0.298)
Strength of Centralization	1.610* (0.600)	2.143* (0.646)	1.684* (0.715)	1.539* (0.582)
Relative Rebel Strength	-0.069 (0.543)	-0.379 (0.561)	0.847 (0.512)	0.616 (0.478)
Independence/Autonomy Movement	4.695* (0.824)	2.029 (1.206)	4.140* (1.674)	1.355 (1.426)
Founded w/ Ideology	0.382 (1.387)	0.870 (1.688)	2.778 (1.701)	0.749 (2.209)
Founded w/ Ethnic Identity	-3.776* (0.988)	-3.162 (1.646)	-3.812* (1.571)	-1.053 (1.125)
External Supporters	0.507* (0.121)	0.305* (0.140)	0.168 (0.113)	0.241 (0.132)
Polyarchy	-0.241 (2.302)	-6.873* (2.657)	1.176 (1.833)	2.398 (1.917)
GDP per capita (thousands)	0.056 (0.076)	0.301* (0.088)	0.019 (0.060)	-0.029 (0.071)
Population (millions)	0.002 (0.001)	0.005* (0.001)	-0.000 (0.001)	0.001 (0.001)
Intensity Level	-0.081 (0.374)	-0.692 (0.433)	0.542* (0.267)	0.736* (0.210)
Constant	-11.039* (2.976)	-8.646* (4.391)	-11.801* (4.074)	-9.782* (3.067)
Region Fixed Effects	✓	✓	✓	✓
Observations	732	751	740	719
Pseudo R^2	0.502	0.413	0.344	0.286

Standard errors in parentheses

Only having non-lootable resources is the baseline across models.

* $p < .05$

Table A3 - Governance Provision with Aggregate Measure of Resources

	(1) M1: Security Institutions	(2) M2: Social Services
Any Natural Resources	1.812* (0.492)	1.448* (0.607)
Territorial Control	0.718* (0.248)	0.681* (0.199)
Strength of Centralization	1.128 (0.644)	1.959* (0.688)
Relative Rebel Strength	0.787 (0.476)	0.653 (0.442)
Independence/Autonomy Movement	3.792* (0.903)	2.104* (0.595)
Founded w/ Ideology	-0.275 (1.607)	0.435 (0.971)
Founded w/ Ethnic Identity	-1.575 (0.957)	-1.996* (0.768)
External Supporters	-0.009 (0.098)	0.322* (0.109)
Polyarchy	-0.424 (2.131)	-1.700 (2.353)
GDP per capita (thousands)	-0.204 (0.155)	0.114 (0.074)
Population (millions)	0.002 (0.001)	0.003* (0.001)
Intensity Level	0.530 (0.290)	0.314 (0.356)
Constant	-7.969* (3.088)	-6.861* (2.773)
Region Fixed Effects	✓	✓
Observations	783	783
Pseudo R^2	0.389	0.366

Standard errors in parentheses

* $p < .05$

Table A4 - Count of Institutions with Aggregate Measure of Resources

	M1: Security Institutions (H1) (Count)	M2: Social Services (H2) (Count)
Any Natural Resources	0.718* (0.286)	0.667* (0.284)
Territorial Control	0.396* (0.103)	0.526* (0.109)
Strength of Centralization	0.474* (0.230)	1.160* (0.284)
Relative Rebel Strength	0.545* (0.223)	0.378 (0.250)
Independence/Autonomy Movement	2.268* (0.459)	1.547* (0.469)
Founded w/ Ideology	-0.190 (1.032)	0.750 (0.696)
Founded w/ Ethnic Identity	-0.649 (0.382)	-1.663* (0.374)
External Supporters	-0.019 (0.054)	0.166* (0.054)
Polyarchy	0.466 (0.748)	-0.877 (1.017)
GDP per capita (thousands)	-0.122 (0.068)	0.058 (0.031)
Population (millions)	0.000 (0.001)	0.001* (0.001)
Intensity Level	0.141 (0.145)	0.006 (0.164)
Constant	-5.516* (1.209)	-4.920* (1.507)
Region Fixed Effects	✓	✓
Log(alpha)	-16.315* (0.166)	-0.298 (0.533)
Observations	783	783
Pseudo R^2	0.283	0.159

Standard errors in parentheses

* $p < .05$

Table A5 - Governance Provision with Year-level Fixed Effects

	M1: Security Institutions (H1)	M2: Social Services (H2)	M3: Social Services (H3)	M4: Security Institutions (H4)
No Natural Resources			0.021 (1.707)	-3.764* (1.017)
Only Non-lootable Resources	3.764* (1.017)	-0.021 (1.707)		
Both Lootable and Non-lootable	1.752* (0.604)	2.101* (0.455)	2.121 (1.588)	-2.012* (0.580)
Territorial Control	0.837* (0.285)	0.748* (0.226)	0.748* (0.226)	0.837* (0.285)
Strength of Centralization	1.591* (0.703)	1.747* (0.575)	1.747* (0.575)	1.591* (0.703)
Relative Rebel Strength	0.971 (0.519)	0.684 (0.434)	0.684 (0.434)	0.971 (0.519)
Independence/Autonomy Movement	4.134* (1.131)	2.157* (0.598)	2.157* (0.598)	4.134* (1.131)
Founded w/ Ideology	0.334 (1.481)	0.121 (1.079)	0.121 (1.079)	0.334 (1.481)
Founded w/ Ethnic Identity	-2.104* (1.070)	-1.805* (0.759)	-1.805* (0.759)	-2.104* (1.070)
External Supporters	-0.043 (0.101)	0.308* (0.111)	0.308* (0.111)	-0.043 (0.101)
Polyarchy	-1.409 (1.835)	-1.601 (2.626)	-1.601 (2.626)	-1.409 (1.835)
GDP per capita (thousands)	-0.122 (0.127)	0.114 (0.101)	0.114 (0.101)	-0.122 (0.127)
Population (millions)	0.002 (0.001)	0.003* (0.001)	0.003* (0.001)	0.002 (0.001)
Intensity Level	0.466 (0.304)	0.494 (0.285)	0.494 (0.285)	0.466 (0.304)
Constant	-9.311* (3.222)	-6.845* (2.487)	-6.866* (2.824)	-5.547 (2.837)
Region Fixed Effects	✓	✓	✓	✓
Year Fixed Effects	✓	✓	✓	✓
Observations	783	783	783	783
Pseudo R^2	0.448	0.406	0.406	0.448

Standard errors in parentheses

No resources is the baseline in Models 1 and 2. Only having non-lootable resources is the baseline in Models 3 and 4.

* $p < .05$

Table A6 - Count of Institutions with Year-level Fixed Effects

	M1: Security Institutions (H1) (Count)	M2: Social Services (H2) (Count)	M3: Social Services (H3) (Count)	M4: Security Institutions (H4) (Count)
No Natural Resources			0.256 (0.913)	-1.424* (0.386)
Only Non-lootable Resources	1.424* (0.386)	-0.256 (0.913)		
Both Lootable and Non-lootable	0.604* (0.295)	0.937* (0.275)	1.192 (0.779)	-0.820* (0.236)
Territorial Control	0.394* (0.099)	0.540* (0.088)	0.540* (0.088)	0.394* (0.099)
Strength of Centralization	0.581* (0.215)	1.034* (0.230)	1.034* (0.230)	0.581* (0.215)
Relative Rebel Strength	0.593* (0.220)	0.378 (0.279)	0.378 (0.279)	0.593* (0.220)
Independence/Autonomy Movement	2.283* (0.464)	1.509* (0.412)	1.509* (0.412)	2.283* (0.464)
Founded w/ Ideology	0.087 (1.006)	0.535 (0.716)	0.535 (0.716)	0.087 (1.006)
Founded w/ Ethnic Identity	-0.796* (0.369)	-1.551* (0.356)	-1.551* (0.356)	-0.796* (0.369)
External Supporters	-0.039 (0.050)	0.158* (0.055)	0.158* (0.055)	-0.039 (0.050)
Polyarchy	0.064 (0.731)	-1.050 (0.920)	-1.050 (0.920)	0.064 (0.731)
GDP per capita (thousands)	-0.087 (0.059)	0.065 (0.034)	0.065 (0.034)	-0.087 (0.059)
Population (millions)	0.000 (0.001)	0.002* (0.001)	0.002* (0.001)	0.000 (0.001)
Intensity Level	0.104 (0.148)	0.089 (0.144)	0.089 (0.144)	0.104 (0.148)
Constant	-5.776* (1.334)	-4.312* (1.402)	-4.568* (1.374)	-4.353* (1.395)
Region Fixed Effects	✓	✓	✓	✓
Year Fixed Effects	✓	✓	✓	✓
Log(alpha)	-16.711* (0.130)	-0.446 (0.588)	-0.446 (0.588)	-15.884* (0.443)
Observations	783	783	783	783
Pseudo R^2	0.300	0.176	0.176	0.300

Standard errors in parentheses

No resources is the baseline in Models 1 and 2. Only having non-lootable resources is the baseline in Models 3 and 4.

* $p < .05$

Table A7 - Governance Provision with Country-level Fixed Effects

	M1: Security Institutions (H1)	M2: Social Services (H2)	M3: Social Services (H3)	M4: Security Institutions (H4)
No Natural Resources			-0.644 (0.683)	-5.200* (0.594)
Only Non-lootable Resources	5.200* (0.594)	0.644 (0.683)		
Both Lootable and Non-lootable	2.549* (0.518)	1.869* (0.661)	1.225* (0.533)	-2.651* (0.573)
Territorial Control	0.720* (0.232)	0.802 (0.538)	0.802 (0.538)	0.720* (0.232)
Strength of Centralization	2.148* (0.559)	5.203* (1.243)	5.203* (1.243)	2.148* (0.559)
Relative Rebel Strength	-0.308 (0.728)	2.373 (1.269)	2.373 (1.269)	-0.308 (0.728)
Independence/Autonomy Movement	6.020* (1.229)	4.760* (2.128)	4.760* (2.128)	6.020* (1.229)
Founded w/ Ideology	-1.555 (1.235)	33.070* (1.899)	33.070* (1.927)	-1.555 (1.235)
Founded w/ Ethnic Identity	-2.268* (0.607)	-6.696* (2.493)	-6.696* (2.493)	-2.268* (0.607)
External Supporters	0.233* (0.107)	0.455* (0.157)	0.455* (0.157)	0.233* (0.107)
Polyarchy	-4.027* (1.714)	3.289 (2.352)	3.289 (2.352)	-4.027* (1.714)
GDP per capita (thousands)	-0.243 (0.134)	-0.234* (0.070)	-0.234* (0.070)	-0.243 (0.134)
Population (millions)	-0.010* (0.003)	-0.005 (0.004)	-0.005 (0.004)	-0.010* (0.003)
Intensity Level	0.428 (0.273)	0.247 (0.390)	0.247 (0.390)	0.428 (0.273)
Constant	-8.314* (3.288)	-49.842* (6.930)	-49.198* (6.846)	-3.114 (3.228)
Region Fixed Effects	✓	✓	✓	✓
Country Fixed Effects	✓	✓	✓	✓
Observations	529	565	565	529
Pseudo R^2	0.459	0.648	0.648	0.459

Standard errors in parentheses

No resources is the baseline in Models 1 and 2. Only having non-lootable resources is the baseline in Models 3 and 4.

* $p < .05$

Table A8 - Count of Institutions with Country-level Fixed Effects

	M1: Security Institutions (H1) (Count)	M2: Social Services (H2) (Count)	M3: Social Services (H3) (Count)	M4: Security Institutions (H4) (Count)
No Natural Resources			0.341 (0.198)	-1.951* (0.300)
Only Non-lootable Resources	1.951* (0.300)	-0.341 (0.198)		
Both Lootable and Non-lootable	0.991* (0.305)	0.485* (0.118)	0.826* (0.197)	-0.959* (0.232)
Territorial Control	0.299* (0.092)	0.232* (0.076)	0.232* (0.076)	0.299* (0.092)
Strength of Centralization	0.741* (0.237)	1.828* (0.187)	1.829* (0.187)	0.741* (0.237)
Relative Rebel Strength	0.483 (0.368)	1.045* (0.206)	1.045* (0.206)	0.483 (0.368)
Independence/Autonomy Movement	3.811* (0.776)	1.663* (0.253)	1.663* (0.253)	3.811* (0.776)
Founded w/ Ideology	0.169 (0.958)	0.271 (0.524)	0.270 (0.524)	0.169 (0.958)
Founded w/ Ethnic Identity	-1.222* (0.275)	-1.584* (0.183)	-1.584* (0.183)	-1.222* (0.275)
External Supporters	0.080 (0.051)	0.141* (0.024)	0.141* (0.024)	0.080 (0.051)
Polyarchy	-0.879 (0.564)	1.588* (0.504)	1.588* (0.504)	-0.879 (0.564)
GDP per capita (thousands)	-0.106 (0.061)	-0.060* (0.013)	-0.060* (0.013)	-0.106 (0.061)
Population (millions)	-0.003* (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.003* (0.001)
Intensity Level	0.120 (0.095)	0.012 (0.066)	0.012 (0.066)	0.120 (0.095)
Constant	-6.965* (2.165)	-6.558* (1.071)	-6.900* (1.039)	-5.015* (2.240)
Region Fixed Effects	✓	✓	✓	✓
Country Fixed Effects	✓	✓	✓	✓
Log(alpha)	-93.633 (.)	-313.311 (.)	-46.463 (.)	-27.715 (.)
Observations	783	783	783	783
Pseudo R^2	0.394	0.363	0.363	0.394

Standard errors in parentheses

No resources is the baseline in Models 1 and 2. Only having non-lootable resources is the baseline in Models 3 and 4.

* $p < .05$

Table A9 - Count of Institutions with Two-Way Fixed Effects

	M1: Security Institutions (H1) (Count)	M2: Social Services (H2) (Count)	M3: Social Services (H3) (Count)	M4: Security Institutions (H4) (Count)
No Natural Resources			-0.455 (0.241)	-0.094 (0.076)
Only Non-lootable Resources	0.094 (0.076)	0.455 (0.241)		
Both Lootable and Non-lootable	0.058 (0.078)	0.744* (0.295)	0.289 (0.162)	-0.037 (0.074)
External Supporters	0.006 (0.017)	0.055 (0.046)	0.055 (0.046)	0.006 (0.017)
Polyarchy	-0.099 (0.650)	2.010 (1.299)	2.010 (1.299)	-0.099 (0.650)
GDP per capita (thousands)	-0.011 (0.015)	-0.131* (0.061)	-0.131* (0.061)	-0.011 (0.015)
Population (millions)	-0.001 (0.001)	0.000 (0.002)	0.000 (0.002)	-0.001 (0.001)
Intensity Level	0.029 (0.048)	0.010 (0.151)	0.010 (0.151)	0.029 (0.048)
Constant	0.533 (0.334)	0.468 (0.919)	0.923 (0.797)	0.628* (0.305)
Observations	855	855	855	855
R^2	0.764	0.843	0.843	0.764

Standard errors in parentheses

No resources is the baseline in Models 1 and 2. Only having non-lootable resources is the baseline in Models 3 and 4.

* $p < .05$

Table A10 - Prob. of Institutions with Two-Way Fixed Effects

	M1: Security Institutions (H1)	M2: Social Services (H2)	M3: Social Services (H2)	M4: Security Institutions (H1)
No Natural Resources			-0.152* (0.066)	-0.111 (0.057)
Only Non-lootable Resources	0.111 (0.057)	0.152* (0.066)		
Both Resources	0.118 (0.066)	0.199* (0.089)	0.046 (0.062)	0.007 (0.061)
External Supporters	0.009 (0.015)	0.007 (0.012)	0.007 (0.012)	0.009 (0.015)
Polyarchy	-0.240 (0.588)	0.202 (0.233)	0.202 (0.233)	-0.240 (0.588)
GDP per capita (thousands)	-0.006 (0.012)	-0.015 (0.012)	-0.015 (0.012)	-0.006 (0.012)
Population (millions)	-0.001 (0.001)	0.000 (0.000)	0.000 (0.000)	-0.001 (0.001)
Intensity Level	0.029 (0.030)	0.005 (0.028)	0.005 (0.028)	0.029 (0.030)
Constant	0.421 (0.293)	0.269 (0.182)	0.422* (0.162)	0.533 (0.280)
Observations	855	855	855	855
R^2	0.769	0.842	0.842	0.769

Standard errors in parentheses

No resources is the baseline in Models 1 and 2. Only having non-lootable resources is the baseline in Models 3 and 4.

* $p < .05$